

Project name	Key users	Problems & needs	Solutions & features	Requirements
AI Learning Environment (working name) A student-designed learning workspace that teaches from the course files already on the student's device.	Students learning from PDFs on desktop and iPad — primary users • Undergraduates working from lecture slides and textbook chapters • Graduate and thesis students working from papers Design partners — students who test explanations, practice and ease of use Evidence so far: student interview with Linda (September 2026)	• Uploading the same files for context at the start of every session interrupts the study session — “it would be smoother if it could see the file I am studying and explain it” (Linda) • Switching between PDF reader, chat assistant, flashcard app and calendar breaks concentration • Answers are not tied to the page being read, so a student cannot easily check them against the source • Understanding a page once is not retention; practice and review are separate manual steps that get skipped until exams • Course files leave the device whenever they are uploaded to a cloud tool	• Point the app at local course folders; original files stay on the device and the student controls folder access • Open a PDF, select a page or passage, and ask by text or voice • Explanation comes back with the page reference it came from • Practice generated from the same passage: flashcards and short quizzes • Planned review scheduled against the student's calendar and exam dates • Desktop workflow first, iPad afterwards	• Read local course folders under explicit, revocable permission; originals stay unchanged on the device • PDF viewer with page and passage selection • AI able to explain a selected passage usefully, at acceptable latency, using free cloud models by default and paid models the student can choose (decided in Workshop 3) • Every explanation cites the page it came from • Flashcard and quiz generation, stored per course • Review scheduler that accepts exam dates and calendar entries • Voice input (later phase) • Scope two people can deliver in one term: one desktop workflow, one course